Rishabh Kushwaha

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PROFESSIONAL SUMMARY

I am a dedicated applied researcher specializing in climate extremes and heat stress, with a focus on Humidex-based diagnostics and their drivers from regional to global scales. Current work includes global future heat stress projections using high-resolution CMIP6 models and India-focused analyses of monthly and seasonal modulation of heat stress by natural climate variability modes (ENSO, IOD, MJO). Experienced in combining ERA5 reanalysis, CMIP6 simulations, and advanced statistical methods (e.g., GEV-based extreme value analysis and teleconnection/regression frameworks) to quantify both mean and extreme heat stress. Research aims to bridge physical climate processes with impact-relevant metrics such as dangerous heat-stress days, supporting risk assessment and climate adaptation planning in vulnerable regions.

RESEARCH INTERESTS

EDUCATIONAL QUALIFICATIONS

Course/Examination	Institution / University	Year of Passing	Percent
Ph. D. in Applied Sciences (Mathematics)	NIT Delhi, New Delhi	Ongoing	7.00 CGPA
Master's in mathematics	Jamia Millia Islamia University, New Delhi	2019	7.00 CGPA
Bachelor of Science	DAV college, Kanpur	2017	55.0 %
Class XII	Up Board	2013	89.0 %
Class X	Up Board	2011	81.0 %

ACHIEVEMENTS & AWARDS

- Qualified JEE Mains (2014)
- Qualified JAM (2017)
- Qualified GATE (2021, 2022)
- Qualified NET (2020, 2021)
- Awarded **Inspire Fellowship scheme.**

DETAILS OF Ph.D. WORK

Thesis Title:	Humidex-Based Heat Stress in a Warming Climate: Global Projections, Natural Climate Variability, and Regional Impacts over India
Supervisor Name and Affiliation:	Dr. Prashant Kumar, Associate Professor, NIT Delhi, India

Brief Description of Ph.D. Work:	My Ph.D. research quantifies present and future heat stress using Humidex as a combined temperature–humidity metric. Globally, I use high-resolution CMIP6 models to project regional Humidex changes and identify future heat-stress hotspots under different SSP scenarios, while over India I use ERA5 and nonstationary extreme-value methods to assess how ENSO, IOD, and MJO modulate monthly mean, extreme Humidex, and dangerous heat-day frequency. Together, this work links large-scale climate variability with human-relevant heat-stress risk to support impact assessment and early-warning strategies.

EMPLOYMENT HISTORY

S. No.	Name of the Post	Institution/University	Period (Year)	Brief description of work
1.	Senior Research Fellow	NIT Delhi	(07-09-2024 to ongoing)	Humidex-Based Heat Stress in a Warming Climate
2.	Junior Research Fellow	NIT Delhi	2 Years, 6 months (16-03-2022 to 06-09-2024)	------

TECHNICAL SKILLS

- **Software's -** MATLAB, Origin, Latex, MS Word, PowerPoint, Excel, Power BI, Mathematica, Grads, Ferret, CDO, ArcGIS.
- **Languages -** MATLAB, Python, C/C++, Fortran, Python.
- **Soft Skills -** Teamwork, Leadership, Event Management, Written & Oral Communication.

PUBLICATIONS

(a) Journal Articles

1. Kushwaha, R., Kumar, P., & Hisaki, Y. (2025). Global future heat stress projections: Regional variations of Humidex changes from high-resolution CMIP6 models. Atmospheric Research, 308, 108367. https://doi.org/10.1016/j.atmosres.2025.108367

(b) Conference Proceedings:

1. Kushwaha, R., and P. Kumar, "Seasonal Influence of ENSO, IOD, and MJO on Heat Index-Derived Heat Stress Across India," presented at SCOPE 2025, IIT Kanpur, November 2025.

REFERENCES

Dr. Prashant Kumar

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Date: 17/Nov/2025 **Rishabh Kushwaha**